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Seasonal Analysis of the Distribution of Urban Traffic Accidents: The Case of İzmir Province

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Abstract – The rapid increase in economic and technological developments in recent years has led to a significant increase in vehicle and pedestrian mobility in cities. While this situation makes daily life easier, it has also brought about some difficulties. Traffic accidents are at the forefront of these problems. Traffic accidents, which are frequently encountered especially in underdeveloped and developing countries, can vary both regionally and seasonally. In this study, it was aimed to analyze on a seasonal basis 26,636 urban traffic accidents that occurred in the city center of İzmir between 2020 and 2023 and were recorded in police reports. In this context, 11 central districts of İzmir were considered separately, the number of urban traffic accidents on a seasonal basis by year were determined and accident density indicators were created. As a result of the analyzes, it was determined that the highest number of traffic accidents between 2020-2023 were generally seen in the autumn (September, October and November) and summer (June, July and August) seasons. Excluding 2020, it was observed that the lowest number of accidents occurred in the winter (December, January and February) season. The findings clearly show that measures and sanctions appropriate to seasonal conditions are very important to increase traffic safety. In addition, it is envisaged that the number of traffic accidents in road networks can be reduced through practices such as information campaigns, road improvements and local warning systems.

Keywords – Traffic accident, Seasonal distribution, Traffic, Regional traffic accidents, Traffic accident density

I. INTRODUCTION

Road traffic, in its simplest definition, encompasses the movements of vehicles and pedestrians on roadways. Since the human factor (driver, passenger, and pedestrian) plays a decisive role in traffic, the risk of errors and accidents is quite high. Traffic accidents, which are among the world's biggest problems, cause approximately 1.5 million deaths and 50 million injuries worldwide each year. This situation also leads to billions of dollars in financial losses on a global scale [1]. Furthermore, the traumatic and psychological issues experienced after accidents negatively affect the mental health of many people, either directly or indirectly, in the following process.

Traffic accidents occur more frequently in underdeveloped and developing countries compared to developed countries [2]. In Turkey, considering the increasing number of vehicles and economic and technological developments, it can be said that road traffic mobility and, consequently, traffic accidents have significantly increased. A detailed examination of road accidents in Turkey reveals that approximately 99% of these accidents are caused by driver, pedestrian, and passenger errors, while only 1% are due to road defects [3]. Moreover, it is known that driver errors are more dominant compared to pedestrian and passenger errors among these accidents [4].

Human errors in traffic can occur for various reasons. These include alcohol and/or drug use, lack of basic traffic knowledge and culture, fatigue and sleep deprivation, excessive stress and tension, and the negative effects of climate and weather conditions on roads and people. When previous studies in the literature are

examined, many studies reveal that climate and weather conditions are important factors in the occurrence of traffic accidents [5]. Some studies emphasize that increasing temperatures and exposure to sunlight, especially in summer months, cause an increase in traffic accidents [6]-[8]. However, it has also been stated that the lowest number of accidents occurs in the winter months [6], [9]. In a study conducted by Farmer and Williams (2005) in the United States, it was highlighted that more accidents occur in the summer and autumn compared to the winter and spring seasons, with the primary reason being the increased vehicle traffic [10]. Spasova (2012), in her study on the city of Sofia, stated that traffic accidents occur most frequently in the autumn and least in the summer [11]. In the study conducted by Shafay et al. (2024) on the city of Hyderabad in Pakistan, it was reported that traffic accidents increased in extreme weather conditions in summer and winter seasons [12]. Similarly, in another study conducted by Eisenberg and Warner (2005) in the United States, it was concluded that despite lower traffic volume in the winter, accident frequency increased [13].

When the studies in the literature are carefully examined, it is clearly seen that the seasonal changes in traffic accidents can vary from country to country and even from region to region. For this reason, it is thought that it would be more appropriate to examine the seasonal distribution of traffic accidents on a smaller scale rather than on a large scale. In this study, it was aimed to analyze the urban traffic accidents that occurred in the city center of İzmir between the years 2020-2023 and recorded in the police reports on a seasonal basis. In this context, 11 central districts of İzmir were considered separately, the number of urban traffic accidents on a seasonal basis according to years were determined and accident density indicators were created.

This study consists of four sections. The second section provides general information about the province of İzmir and the central districts examined in the study. The third section presents a detailed seasonal analysis of traffic accidents on a district basis. In the final section, the results obtained from the study are evaluated and interpreted. Additionally, future recommendations and thoughts are shared.

II. STUDY AREA: CENTRAL DISTRICTS OF İZMİR PROVINCE

Located in the Aegean Region of Turkey, İzmir is the third largest city with the highest population density after İstanbul and Ankara. In addition to being an important city in historical, touristic and socio-cultural terms, it also has a strategic importance in economic terms [14]. İzmir, with a surface area of 11,892 km², is located at the southeastern end of İzmir Bay. The locations of İzmir and its districts on the map are shown in Fig. 1 [15].

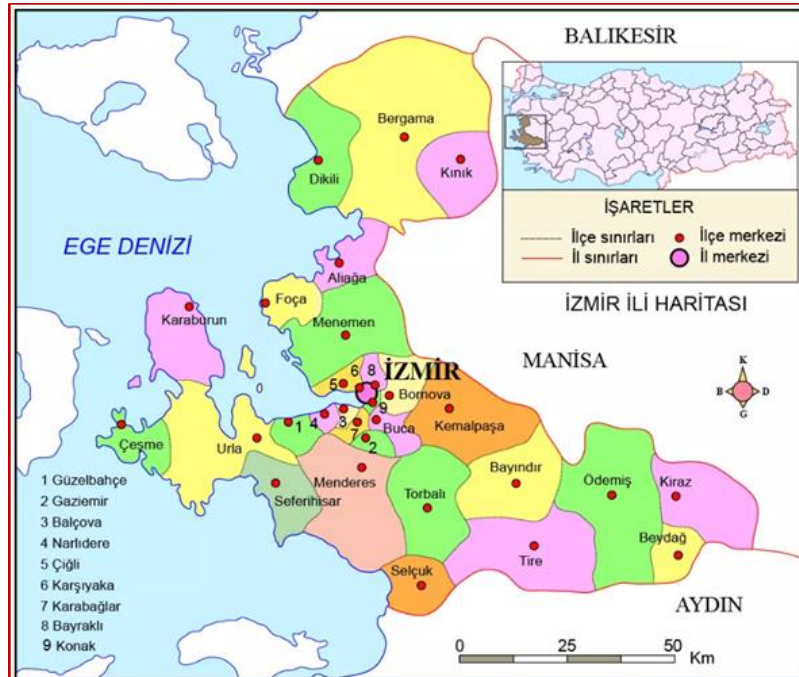


Fig. 1 Location of İzmir and its districts on the map [15]

As seen in Figure 1, İzmir consists of 30 districts in total; 11 of them are the center and 19 are other districts. As of the end of 2023, the total population of İzmir is approximately 4.5 million. There are approximately 380 people per km² in the province. The central district with the highest population density is Konak. The three largest central districts in terms of population are Buca, Karabağlar and Bornova. The central district with the least population is Güzelbahçe. Total population information of the central districts is presented in Table 1 [14].

Table 1. Surface area and population information of central districts of İzmir province [14]

Central Districts of İzmir Province	Surface Area (km ²)	Population (person)
Bağcıva	16	77,831
Bayraklı	30	300,949
Bornova	220	447,553
Buca	178	523,487
Çiğli	139	215,172
Gazimir	70	137,768
Güzelbahçe	77	38,044
Karabağlar	89	476,500
Karşıyaka	51	341,857
Konak	24	327,300
Narlıdere	50	62,148

İzmir has a Mediterranean climate. Summers are hot and dry, and winters are mild and rainy. Snowfall is very rare in the central districts. Temperatures are above 30°C for about a third of the year, and the number of days when the temperature drops below 0°C is very few [14].

Since the historical and touristic structures and summer resorts of İzmir are located mainly in the districts of Konak, Karşıyaka and Güzelbahçe, the traffic density in these districts is quite high, especially in the summer season. In addition, serious traffic density and problems are observed almost in every season of the year in the districts of Karabağlar, Buca and Bornova, where settlements are dense and industrial zones are located.

III. SEASONAL ANALYSIS OF TRAFFIC ACCIDENTS

In this part of the study, urban traffic accidents were analyzed on a seasonal basis using the data of 26,636 traffic accidents that occurred in the police responsibility area between 2020 and 2023 in the central districts of İzmir province, provided by the Traffic Directorate of the General Directorate of Security of the Ministry of Internal Affairs of the Ministry of Internal Affairs of the Republic of Turkey. The total number of accidents that occurred in the central districts of İzmir between the years 2020-2023 and reflected in the police reports are presented in Table 2.

Table 2. Total number of accidents that occurred in the central districts of İzmir between 2020-2023 and were recorded in police reports

Central Districts of İzmir Province	2020	2021	2022	2023	2020-2023
Bağcıva	147	203	247	272	869
Bayraklı	272	511	580	648	2,011
Bornova	852	1,308	1,404	1,525	5,089
Buca	751	1,062	1,178	1,198	4,189
Çiğli	265	407	428	475	1,575
Gazimir	227	314	405	433	1,379
Güzelbahçe	71	111	136	174	492

Karabağlar	303	827	818	910	2,858
Karşıyaka	537	604	704	685	2,530
Konak	1,170	1,152	1,405	1,329	5,056
Narlidere	120	168	156	144	588

In this section, firstly, the existing accident data were evaluated separately and the monthly urban accident numbers were determined for all central districts. The number of urban traffic accidents and accident density indicators on a seasonal basis for the years 2020-2023 are presented in Fig. 2.

	2020					2021			
	Winter	Spring	Summer	Autumn		Winter	Spring	Summer	Autumn
Balçova	39	22	44	42	Balçova	38	35	67	63
Bayraklı	70	40	49	113	Bayraklı	88	122	148	153
Bornova	219	125	241	267	Bornova	260	307	366	375
Buca	195	126	231	199	Buca	205	248	315	294
Çiğli	61	47	70	87	Çiğli	71	92	114	130
Gaziemir	63	33	72	59	Gaziemir	46	81	88	99
Güzelbahçe	9	14	23	25	Güzelbahçe	22	20	37	32
Karabağlar	71	44	42	146	Karabağlar	174	182	224	247
Karşıyaka	128	108	162	139	Karşıyaka	125	125	157	197
Konak	282	223	367	298	Konak	232	251	334	335
Narlidere	21	18	42	39	Narlidere	26	34	71	37

	2022					2023			
	Winter	Spring	Summer	Autumn		Winter	Spring	Summer	Autumn
Balçova	50	63	60	74	Balçova	55	66	72	79
Bayraklı	131	140	141	168	Bayraklı	121	164	175	188
Bornova	361	329	351	363	Bornova	323	394	392	416
Buca	268	306	285	319	Buca	276	302	286	334
Çiğli	87	123	99	119	Çiğli	108	102	143	122
Gaziemir	76	104	112	113	Gaziemir	99	101	105	128
Güzelbahçe	28	32	38	38	Güzelbahçe	35	30	59	50
Karabağlar	175	217	198	228	Karabağlar	187	263	235	225
Karşıyaka	162	209	151	182	Karşıyaka	152	176	164	193
Konak	300	410	349	346	Konak	294	365	333	337
Narlidere	28	44	40	44	Narlidere	30	35	42	37

	2020 - 2023			
	Winter	Spring	Summer	Autumn
Balçova	182	186	243	258
Bayraklı	410	466	513	622
Bornova	1163	1155	1350	1421
Buca	944	982	1117	1146
Çiğli	327	364	426	458
Gaziemir	284	319	377	399
Güzelbahçe	94	96	157	145
Karabağlar	607	706	699	846
Karşıyaka	567	618	634	711
Konak	1108	1249	1383	1316
Narlidere	105	131	195	157

Fig. 2 Number of urban traffic accidents and accident density indicators on a seasonal basis for the years 2020-2023

As seen in Figure 2, the highest number of traffic accidents between 2020 and 2023 were generally seen in the autumn (September, October and November) and summer (June, July and August) seasons. However, excluding 2020, it was observed that the lowest number of accidents occurred in the winter season (December, January and February). Although these results are consistent with some studies in the literature [6]-[10], they are not consistent with some studies [11]-[13]. This is an important indicator that the seasonal distribution of accidents may differ from region to region.

Following the declaration of COVID-19 as a "global pandemic" by the World Health Organization (WHO) on March 11, 2020, global measures have also been implemented in our country. In particular, in March, April and May 2020, the Republic of Turkey took various security measures and imposed a curfew

for many days [16]. Therefore, it is not considered an unexpected result that the lowest number of accidents in 2020 was observed in the spring season (March, April and May).

When looking at the total number of accidents between 2020 and 2023, it can be said that urban traffic accidents in all central districts are generally concentrated in autumn, summer, spring and winter months, respectively. Fig. 3 shows seasonal accident density rates by year in detail.

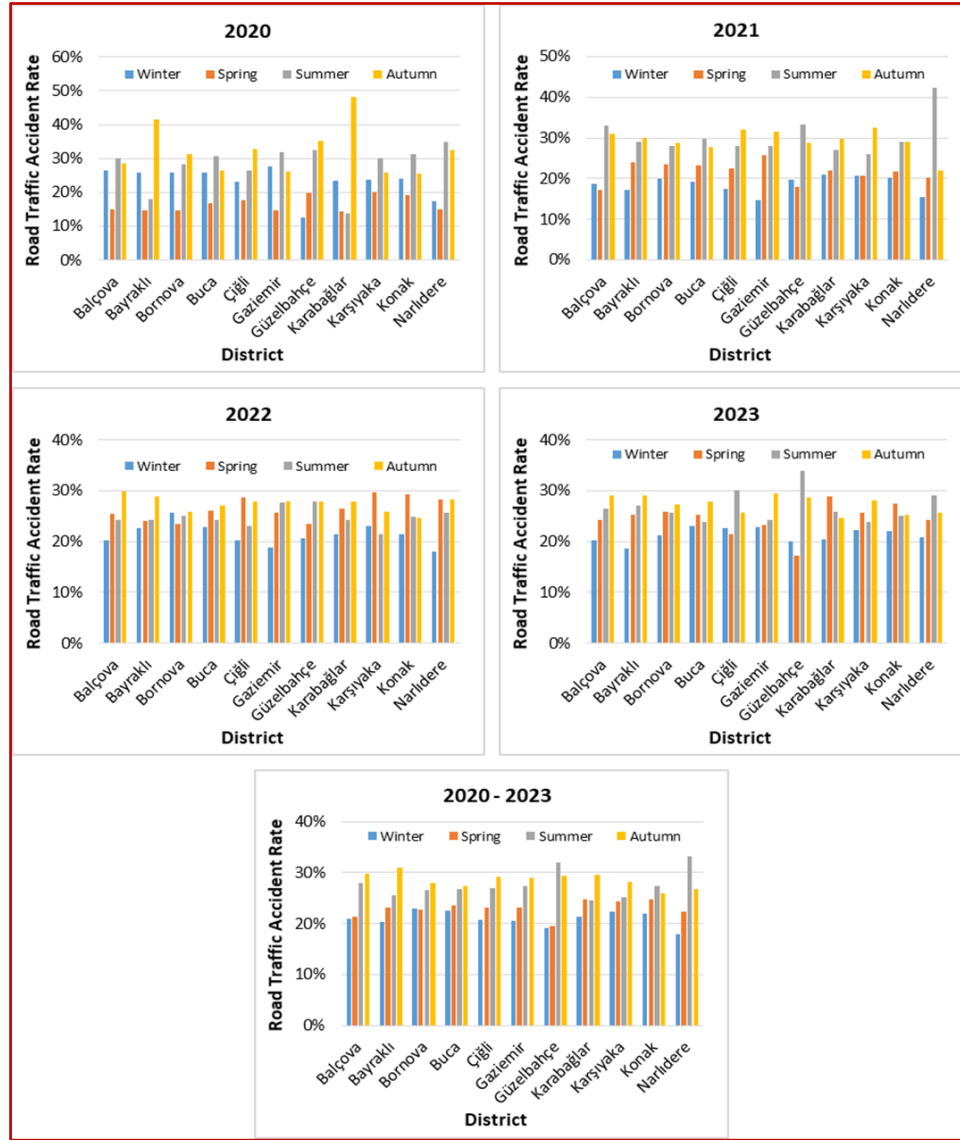


Fig. 3. District-based seasonal accident density rates by year

In the next stage of the study, it was aimed to evaluate the accident density according to years, regardless of the districts. In this context, the seasonal accident numbers were calculated separately on a yearly basis. The number of traffic accidents and accident density indicators obtained as a result of the calculations are presented in Fig. 4.

	Winter	Spring	Summer	Autumn
2020	1158	800	1343	1414
2021	1287	1497	1921	1962
2022	1666	1977	1824	1994
2023	1680	1998	2006	2109
2020-2023	5791	6272	7094	7479

Fig. 4 Year-based number of traffic accidents and accident density indicators for the central districts of İzmir province

In Fig. 4, it is clearly seen that the highest number of traffic accidents occurred in the autumn season in the central districts of İzmir province between 2020 and 2023. Autumn is followed by summer. The lowest number of accidents was generally recorded in the winter season (except for 2020). The proportional distribution of seasonal accident numbers by year is presented graphically in Fig. 5.

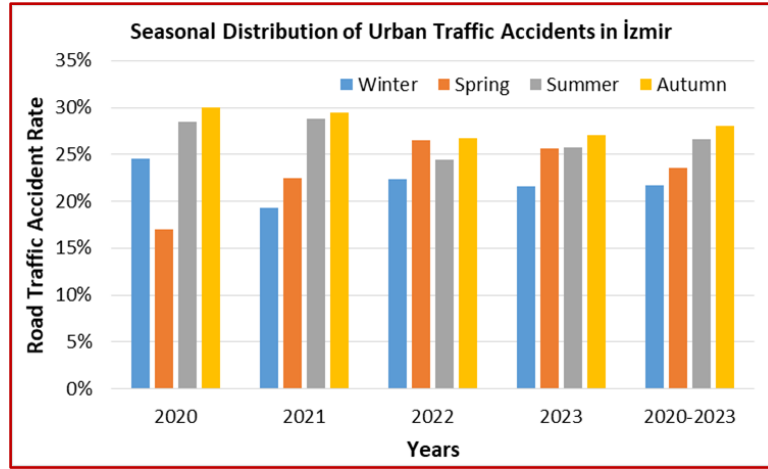


Fig. 5 Proportional distribution of year-based seasonal accident numbers for the central districts of İzmir province

The following evaluations can be made for the seasonal distribution of accidents occurring in the central districts of İzmir province between 2020-2023:

- In 2020, approximately 30% of the 4,715 accidents recorded in police reports occurred in the autumn season, 28.5% in the summer, 24.6% in the winter, and 17% in the spring.
- In 2021, approximately 29.4% of the 6,667 accidents recorded in police reports occurred in the autumn season, 28.8% in the summer, 22.5% in the spring, and 19.3% in the winter.
- In 2022, approximately 26.7% of the 7,461 accidents recorded in police reports occurred in the autumn season, 26.5% in the spring, 24.4% in the summer, and 22.3% in the winter.
- In 2023, approximately 27.1% of the 7,793 accidents recorded in police reports occurred in the autumn season, 25.7% in the summer, 25.6% in the spring, and 21.6% in the winter.
- Approximately 28.1% of the total 26,636 accidents recorded in police reports between 2020 and 2023 occurred in the autumn season, 26.6% in the summer, 23.5% in the spring, and 21.7% in the winter.

IV. CONCLUSIONS AND FUTURE WORKS

This study aims to analyze the urban traffic accidents that occurred in the city center of İzmir province between 2020-2023 and recorded in police reports seasonally. In this context, the data of 26,636 traffic accidents that occurred in the police responsibility area in the central districts of İzmir province between 2020-2023, provided by the Traffic Presidency of the General Directorate of Security of the Ministry of Internal Affairs of the Republic of Turkey, was used. The number of urban traffic accidents was determined on a seasonal basis according to years and accident density indicators were created.

According to the analysis results, it was determined that the highest number of traffic accidents between 2020 and 2023 generally occurred in the Autumn (September, October, November) and Summer (June, July, August) seasons. However, excluding 2020, it was concluded that the lowest number of accidents was seen in the winter (December, January, February) season. It was determined that the least number of traffic accidents in 2020 occurred in the spring (March, April, May) season. It can be said that this situation is due to the curfews implemented by the Republic of Turkey since March 2020. In this context, it is thought that COVID-19 has made a positive contribution to traffic safety.

Since İzmir is a touristic center, it is expected that there will be an increase in urban traffic and therefore traffic accidents, especially in the summer months. It is known that the negative effects of the sweltering summer temperatures on people (reduced physical and mental performance) are also an important factor that increases traffic accidents. In addition, since İzmir has a Mediterranean climate, it is not surprising that

factors such as rain, fog and reduced visibility in the fall increase the number of accidents. It is thought that the fact that road users (drivers and pedestrians) are more careful and cautious in traffic, especially in the winter season, contributes significantly to the general decrease in traffic accidents. As a result, when traffic accidents in Izmir are examined seasonally, it is seen that accidents increase in the fall and summer months, while accidents decrease in the winter months due to careful driving and cautious behaviour. These findings clearly emphasize the importance of measures and sanctions appropriate to seasonal conditions to increase traffic safety. Information campaigns, road improvements and local warning systems are considered to be important steps towards reducing traffic accidents.

In the scope of this study, the seasonal distribution of traffic accidents was examined, but the factors that caused the accidents were not addressed. In future studies, it is aimed to investigate the factors that caused the accidents separately and to explain them in more detail through statistical tests. Thus, it will be possible to better understand the reasons for the differences in seasonal distributions.

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